

KABHILAN V. S

Student

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About Me

Aspiring Software Engineer pursuing an Integrated M.Sc. in Software Systems with hands-on experience in full-stack web and mobile application development, AI-powered solutions, and real-world software projects. Passionate about learning emerging technologies and building scalable, user-centric applications that solve practical problems.

Projects

EUREKA – AI-Powered Semantic Job Role Mapping (STATATHON 2025-26 Grand Finale)

- Developed an AI-powered semantic occupation mapping system that converts free-text and voice-based job descriptions into standardized NCO-2015 occupation codes using semantic search, vector embeddings, and AI-assisted validation. Implemented backend APIs, benchmarking, and intelligent search optimization to improve occupation matching accuracy.
- Tech Stack: Python, Flask, PostgreSQL, FAISS, Sentence Transformers, Bootstrap

A1 Water Tech (Real Client Project)

- Developed a complete website and mobile application for a water supply business by working directly with evolving client requirements. Designed and implemented frontend interfaces, backend services, database connectivity, and business workflows while continuously refining the application based on customer feedback.
- Tech Stack: React, Flutter, Flask, Node.js, Firebase, MySQL

API Monitoring & Downtime Alert Extension (Hackwave 3.0.1)

- Designed and developed a real-time API monitoring platform to track uptime, response time, latency, and service availability. Automated health checks, downtime detection, and centralized monitoring dashboards to improve system reliability and service monitoring.
- Tech Stack: Python, Flask, REST APIs, MySQL

Custom AI Agent

- Built a prompt-driven AI agent framework capable of dynamically generating and executing task-specific workflows from natural language prompts. Explored LLM integration, prompt engineering, and intelligent automation through practical AI development.
- Tech Stack: Python, Ollama, LLMs, AI Agents

FindinFront (Ongoing)

- Developing an AI-assisted mobile application for visually impaired users that provides real-time object awareness through voice guidance and vibration feedback, improving accessibility and independent navigation.
- Tech Stack: Flutter, Firebase, OpenCV, Python

PoachDetect – Animal Poaching Detection System

- Designed a computer vision-based wildlife monitoring system capable of detecting suspicious human activity in protected forest areas using deep learning and image processing techniques to support anti-poaching operations.
- Tech Stack: Python, OpenCV, PyTorch

AgriTrack

- Developed a smart farming management system that helps farmers manage crop planning, agricultural activities, and farm records using structured data analysis and digital management features.
- Tech Stack: React, Flask, MySQL

Attendo

- Developed a digital attendance management system to simplify attendance tracking, record management, and user administration through an intuitive interface with secure backend and database integration.
- Tech Stack: Flutter, Firebase, Flask

V-Learn

- Designed and developed a learning management platform that enables students to access study materials, share educational resources, and improve collaborative learning through a centralized digital environment.
- Tech Stack: React, Node.js, MongoDB

DirectFarm

- Developed a digital platform connecting farmers directly with consumers to improve market accessibility, product management, and transparent agricultural trading.
- Tech Stack: React, Flask, MySQL

Community Guardian (GHCI 2025 Hackathon)

- Developed a Healthcare API Monitoring System focused on improving the reliability, security, and availability of healthcare services through automated API monitoring, health analysis, and intelligent alerting. Selected for Round 2 of the GHCI 2025 Hackathon.
- Tech Stack: Python, Flask, REST APIs, AI Monitoring

Tourist Application (Odoo × SNS Coimbatore Hackathon 2026)

- Led the development of a smart tourism platform designed to assist travelers with destination discovery, trip planning, and location-based recommendations. The project was selected for Round 2 of the hackathon.
- Tech Stack: Flutter, Firebase, Google Maps API, Node.js

Area of Interest

- Software Engineering
- Full-Stack Development
- Artificial Intelligence & Machine Learning
- AI Agent Development
- Backend Systems & REST APIs
- Mobile Application Development
- Semantic Search & Information Retrieval
- Computer Vision
- Cloud Technologies
- Scalable Software Architecture

Leadership & Responsibilities

Executive Head – Coding Club (Department of Computer Technology - PG)

- Lead and coordinate technical activities, coding initiatives, and student engagement programs within the department.
- Mentor students, support collaborative learning, and encourage participation in hackathons and technical competitions.

Team Leader

- Hackware'24 – Led the development of the Coding Club Website, receiving the Appreciation Award.
- Project Expo – Led the DirectFarm project presentation and coordinated team development.
- E-Horyzon – Led the development of the Attendance Management Application.
- KPIT Sparkle – Led the Healthcare API Monitoring & Downtime Alert Extension project.
- AI for Bharat Hackathon – Led the development of an AI-powered solution focused on learning and developer productivity.
- GHCI 2025 Hackathon – Led the Community Guardian project, a Healthcare API Monitoring System, selected for Round 2.
- Odox × SNS Coimbatore Hackathon 2026 – Led the Tourist Application project, selected for Round 2.

Event Coordinator

- Hackwave – Coordinated technical events and managed event execution.
- Code Carnival – Organized coding competitions and student engagement activities.

UI/UX Coordinator

- Quantum Fest – Designed and coordinated user interface and user experience elements for the event.
- MarkUs 2K26 – Managed UI/UX design and development activities for the department event.

Executive Member

- Coding Club (Second Year)
- Contributed to organizing technical events, coding activities, and collaborative student initiatives before becoming Executive Head.

ACHIEVEMENTS & AWARDS

Second Prize – STATATHON 2025–26 Grand Finale (National Level)

- AI-Powered Semantic Job Role Mapping for NCO-2015.

First Prize – KEC Hackathon 2024

- Game Development using Pygame.

First Prize – Proof of Concept 2025

- Animal Poaching Detection using Deep Learning.

First Prize – Quantum Fest 2K25

- Crypto Analysis with Ransomware Detection.

Second Prize – Ideathon 2025

- Animal Poaching Detection using Deep Learning.

Third Prize – Hackwave 3.0.1

- API Monitoring & Downtime Alert Extension.

Appreciation Award – Hackware'24

- Team Leader – Coding Club Website.

First Prize – Video Editing Competition.

Second Prize – Algo Arena (DSA Challenge).

CERTIFICATIONS

- Cloud Computing – Infosys Springboard
- Virtualization and Cloud Computing – Infosys Springboard
- Step-by-Step Machine Learning with Python – Infosys Springboard
- Data Mining Projects with R – Infosys Springboard
- Learning Data Mining with R – Infosys Springboard
- Introduction to Agile Testing – Infosys Springboard
- Testing Tools, Selenium with Java & Python & API Testing – NASSCOM FutureSkills Prime

Languages

- Tamil – Native Proficiency
- English – Professional Working Proficiency

Education

Kongu Engineering College (Autonomous), Perundurai

Integrated Master of Science (M.Sc.) – Software Systems

2023 – Present

Current CGPA: 7.80 / 10.0

Bharathi Vidya Bhavan Matriculation Higher Secondary School, Thindal

Higher Secondary Certificate (HSC) – Mathematics & Computer Science

2022 – 2023

Percentage: 65.16%

Green Garden Matriculation Higher Secondary School, Perundurai RS

Secondary School Leaving Certificate (SSLC)

2020 – 2021

NATIONAL HACKATHONS & SELECTIONS

GHCI 2025 Hackathon

- Team Leader | Community Guardian – Healthcare API Monitoring System
- Selected for Round 2.

Odoo × SNS Coimbatore Hackathon 2026

- Team Leader | Tourist Application
- Selected for Round 2.

AIM'26 Hackathon

- Custom AI Agent.

AI for Bharat Hackathon

- Team Leader | AI for Learning & Developer Productivity.

KPIT Sparkle

- Team Leader | API Monitoring & Downtime Alert Extension.

Odoo × NMIT Hackathon 2025

- SynergySphere – Advanced Team Collaboration Platform.

START-A-THON'25

- Plastic Waste & Clean Water Detection.

Smart India Hackathon (SIH)

- Developed a project solution and participated in the internal selection process.

TECHNICAL EVENTS & PRESENTATIONS

Project Presentations

- Project Expo – DirectFarm (Team Leader)
- E-Horyzon – Attendance Management Application (Team Leader)
- Hacknovate'25 – KEC Study Hub
- Hackwave 2.0 – Machine Learning for Stock Price Forecasting
- Internal Hackathon – Improving Open Source Software Security through Fuzzing
- NASSCOM – Emergency Drone System
- Quantum Fest – Emergency Drone System (Paper Presentation)

Technical Competitions

- **Cryptera 2025**
 - CSS Battle
 - UI Design
 - Code in Chaos
 - Prompt Engineering
- **Quantum Fest**
 - Tech Quiz
- **Login 2K24**
 - AlgoCode Quiz
 - Picture Perfect (Photography)
- **Code-o-Fiesta**

TECHNICAL SKILLS

Programming Languages

- Python
- Java
- JavaScript
- SQL
- C
- C++
- PHP (Basic)

Frontend Development

- React
- HTML5
- CSS3
- Bootstrap
- Responsive Web Design

Mobile Application Development

- Flutter
- React Native
- Kivy (Basic)

Backend Development

- Flask
- Node.js
- REST API Development
- API Integration

Database Management

- PostgreSQL
- MongoDB
- MySQL
- Firebase
- Database Design

Artificial Intelligence & Machine Learning

- PyTorch
- OpenCV
- Semantic Search
- AI Agent Development
- Prompt Engineering
- Vector Embeddings
- Ollama
- Computer Vision
- Large Language Model (LLM) Integration

Cloud & DevOps

- AWS
- Firebase
- Docker (Basic)

Development Tools

- Git
- GitHub
- Visual Studio Code
- Android Studio
- Postman

Software Engineering

- Object-Oriented Programming (OOP)
- Data Structures & Algorithms
- Software Development Life Cycle (SDLC)
- System Design
- Requirement Analysis
- Debugging & Testing